

AI Project Report ? 2026-09-02

Credit Default Prediction with Logistic Regression

Final session report

1. Daily Executive Summary

The session produced and documented a logistic-regression workflow for predicting the binary Defaulted outcome from the workspace dataset. Applicant_ID was excluded as a direct identifier. The final notebook uses median imputation and standardization inside repeated stratified cross-validation, compares the model with a prior-probability baseline, and reports probability, discrimination, threshold, and confusion-matrix metrics.

The user executed and saved the final notebook. Saved results show perfect aggregated out-of-fold classification and a mean repeated-fold accuracy of 0.997. These results are preliminary: the dataset contains only 50 observations, and no external validation, formal calibration assessment, or deployment review was performed. Codex did not execute the model.

2. Report Metadata

Field	Value
Report date and time	2026-09-02 13:18 +08:00 (Asia/Manila)
Repository	C:\Jan Bush Files\Coding_Miscellaneous\GitHub\VibeCodingTest_v2
Branch	main
Relevant HEAD	4af81dc0c09338510caaf946bdcfaefd1888c708
Primary implementation agent	Codex; exact model label not established by workspace evidence
Secondary reviewer	GitHub Copilot; model unknown; no usable review provided
Batches	BATCH_02-09-2026_02_V1, BATCH_02-09-2026_03_V1, BATCH_02-09-2026_04_V1
Git state before report generation	GPT_Codes.md modified by the final session-summary append; no Codex Git mutation

3. Objectives and Work Completed

- ? Identified the predictive objective as binary default classification.
- ? Rejected ordinary least-squares regression as the primary model because Defaulted is binary.
- ? Selected L2-regularized logistic regression.
- ? Excluded Applicant_ID from modeling and displayed predictions.
- ? Created and uploaded a reviewable Jupyter notebook.

- ? Implemented leakage-safe median imputation and standardization inside validation folds.
- ? Implemented repeated stratified 5-fold cross-validation with 20 repetitions.
- ? Compared logistic regression with a prior-probability dummy baseline.
- ? Added ROC-AUC, log loss, accuracy, balanced accuracy, precision, sensitivity, specificity, F1, negative predictive value, and confusion-matrix counts.
- ? Maintained the append-only implementation ledger and documented corrections, approvals, deviations, and session closure.

4. Files Created or Changed

File	Authorship / role	Session treatment
credit_risk_screening.csv	Human-provided source data	Read for analysis; not modified by Codex
credit_default_logistic_regression.ipynb	Codex initial artifact and approved edits; human execution and cleanup	Created, uploaded, edited, then executed and saved by the user
GPT_Codes.md	Codex append-only ledger; included in user commits	Created, uploaded, and appended under standing permission
COPILOT_REVIEWS.md	Created by user	Contains a statement that Copilot could not provide useful review

5. Important Decisions and Reasons

- ? Logistic rather than linear regression: the target is binary, so logistic regression constrains fitted probabilities and matches the classification objective.
- ? Identifier exclusion: Applicant_ID has no justified predictive role and would create memorization and privacy risk.
- ? Fold-contained preprocessing: imputation and scaling are fitted only on each training fold to prevent leakage.
- ? Repeated stratified validation: the dataset is too small for a stable single holdout, while stratification preserves class support.
- ? Multiple metrics: accuracy alone cannot describe probability quality or false-positive and false-negative behavior.
- ? Saved outputs preserved during approved edits: user notebook state was treated as user work; edited-code outputs were not represented as newly verified until the user reran the notebook.

6. Verification Results

USER-REPORTED RUNTIME ? saved notebook outputs

Metric	Logistic regression	Prior baseline
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ROC-AUC	1.000	0.500
Log loss	0.088	0.693
Accuracy	1.000	0.500
Balanced accuracy	1.000	0.500
Precision	1.000	0.500
Sensitivity	1.000	1.000
Specificity	1.000	0.000
F1	1.000	0.667
Negative predictive value	1.000	Not defined for baseline predictions
Confusion counts	TN 25; FP 0; FN 0; TP 25	TN 0; FP 25; FN 0; TP 25

The saved repeated-fold summary reports mean accuracy 0.997 with standard deviation 0.017 for logistic regression. Saved final-fit standardized coefficients are -1.759 for Annual_Income and 1.910 for Debt_to_Income_Ratio, with exploratory odds ratios of 0.172 and 6.751 per one-standard-deviation increase.

STATIC ? Codex verification

- ? Notebook JSON and Python syntax were inspected without executing notebook cells.
- ? The approved metrics were present in fold-level and aggregated evaluation paths.
- ? The source dataset was preserved.

AI-VERIFIED RUNTIME applied only to the approved upload operations and their checksum checks, not to model performance.

7. Copilot?Codex Reconciliation and Dispositions

No compliant Copilot advisory review or material recommendation was available.

COPILOT_REVIEWS.md contains only the user statement: Copilot was unable to provide useful review.

- ? Disposition: USER-DENOUNCED. The Copilot channel did not influence implementation.
- ? No ACCEPT, ACCEPT WITH CHANGES, REJECT, or DEFER disposition was required because there was no material Copilot recommendation to evaluate.

8. Data-Handling Record

- ? Active mode: ANALYSIS_APPROVED.
- ? Source dataset: credit_risk_screening.csv; treated as immutable.
- ? Likely sensitive categories: applicant identifier and financial attributes.
- ? Applicant_ID was excluded from modeling and persistent result displays.
- ? No row-level applicant values, direct identifiers, or secrets are reproduced in this report.
- ? No derived dataset or serialized model was written automatically.

9. Problems, Conflicts, Risks, and Limitations

- ? Initial methodological conflict: ordinary linear regression was unsuitable for the binary target; logistic regression resolved the conflict.
- ? Notebook-buffer conflict: an unsaved human cell appeared during an approved disk edit. Codex stopped and preserved it; the user later confirmed and removed the faulty cell.
- ? Sample size: only 50 observations, despite balanced classes.
- ? Perfect aggregated predictions may reflect unusually strong separation in this small test dataset and do not establish external generalization.
- ? No external validation, formal calibration analysis, fairness assessment, causal inference, or deployment governance review was completed.
- ? Repeated-fold percentiles are descriptive and are not formal confidence intervals.
- ? Saved output reports a scikit-learn deprecation warning for the explicit penalty='l2' argument; future compatibility cleanup may be needed.

10. Pending Work and Recommended Next Steps

No work remains necessary for the stated session objective. Optional future work:

- ? Validate on substantially more representative external data.
- ? Assess calibration and choose a classification threshold using explicit decision costs.
- ? Investigate whether perfect separation persists and whether data provenance or leakage concerns exist outside the implemented pipeline.
- ? Update the LogisticRegression configuration for future scikit-learn compatibility.
- ? Perform formal fairness, governance, and domain review before any consequential use.

11. Provenance

Source	SHA-256 / identifier
GPT_Codes.md before report generation	b47b98740e5babaf4ab5149e758743d9452cd6eb061850c7744a41e76423f79
COPILLOT_REVIEWS.md	32e7e0da872729f46f7f0cca8c1f9eade64b199dc25097b1a3a8ef15896b9636
credit_default_logistic_regression.ipynb	31b512df935223c27140afe8b51efe9f0558d6b1f407fadba1bf9a83fb2ffa29
credit_risk_screening.csv	6d204b77c878d48232197fe9e2a8fcae47baed47749ada91b865eb1ad575f038
Git HEAD	4af81dc0c09338510caaf946bdcfaefd1888c708

Authoritative sources remain GPT_Codes.md, COPILLOT_REVIEWS.md, the notebook, the immutable source dataset, and Git history. This Word report is a derived presentation layer. Report generation occurred only in an isolated temporary delivery location and did not modify the source Markdown records, notebook, dataset, Git index, commit history, or remote repository.